

Ryaan Farrukh

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EXPERIENCE

Software Engineer Intern

Jan 2026 – Apr 2026

TD Bank

Toronto, Ontario

- Built backend document workflows in TD's **Document Ingestion** platform using **Tungsten TotalAgility**.
- Developed workflow logic in **C#/.NET** across classification, extraction, and export pipeline stages.
- Debugged **extraction and routing failures** using log analysis, step-level tracing, and pre-release validation.
- Designed and delivered **4 end-to-end POCs** using TTA + GenAI, presenting results to **business stakeholders**.
- Collaborated in an **Agile** team while remediating **300+ vulnerabilities** and supporting COTS deployments.

Software Developer Intern

Jan 2025 – Apr 2025

Unity Fitness

Toronto, Ontario

- **Boosted efficiency by 40%** by developing **JavaScript automation workflows** reducing manual effort.
- **Enhanced platform responsiveness and UX** by building interactive **UI components in JavaScript**.
- **Cut MongoDB response times 60%** via query optimization, ensuring **99.9% accuracy on 10K+ records**.
- **Launched a React/Node.js/MongoDB app** with **GPT-3 integration**, boosting scalability and features.
- **Improved reliability and reduced bugs** by conducting **API testing and validation in Postman**.

PROJECTS

QueueForge | Go, Redis, PostgreSQL, Docker 🔗

Apr 2026

- Engineered **async job queue in Go** using **Redis LIST/ZSET** and **Postgres** for durable job state persistence.
- Built **concurrent worker pool** with **goroutines** to process distributed jobs across API and worker services.
- Implemented **retry logic** with **exponential backoff**, delayed queues, and **dead-letter handling** for failures.
- Designed **REST API** for **job submission**, status tracking, and lifecycle control across worker service nodes.
- Containerized **API and workers** with **Docker Compose**, enabling horizontal scaling across service instances.

Virtual File System | C++ 🔗

Mar 2025

- Developed a **virtual file system in C++**, simulating binary storage with block-level allocation.
- Implemented **UNIX-style commands** (`mkdir`, `touch`, `cd`, `rm`, `write`) for file management.
- Designed a **metadata structure** enabling hierarchical storage and recursive traversal of deep trees.
- Added **UID login and UNIX permission bits (rwx)** to enforce secure and controlled system access.
- Stress-tested with workloads, ensuring stability and safe **memory handling** during large operations.

MarketVision | Go, Node.js, Next.js, MongoDB 🔗

Feb 2025

- Developed a **Go backend with Gin** to process AI image requests, returning **structured product data**.
- Built a responsive **Next.js + Tailwind CSS frontend**, delivering seamless UX and **real-time AI results**.
- Implemented **JWT authentication with bcrypt/MongoDB**, securing accounts and supporting **15+ users**.
- Designed an efficient **MongoDB schema** for prompts, responses, and base64 images with fast retrieval.
- Optimized **API handling and UI**, boosting performance, scalability, and improving **user retention**.

SKILLS

Languages: C/C++, Python, Go, Java, C#/.NET, JavaScript, HTML/CSS, SQL

Technologies: Git, Jira, Docker, AWS, MongoDB, PostgreSQL, Bash, GitHub Actions, Jest, Cypress

Frameworks: React.js, Express.js, Node.js, Next.js, Tailwind, Flask

EDUCATION

Computer Programming & Analysis Co-op

Seneca Polytechnic

Toronto, Ontario

Jan 2024 – Apr 2027